

Human–AI Collaborative Triage in Conversational Social Engineering Detection

Intern/Co-op/Directed Research

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Approval

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Declaration

It is to declare that this project is our original work. No part of this work has been submitted elsewhere, partially or entirely, for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor.
Relevant previous works presented in this report have been adequately acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

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Last but not least, we owe our deepest gratitude to our parents for their sacrifices in reaching this stage of life and for their work on this project.

Abstract

Scammers are now using advanced tactics. They don't send malicious messages immediately; instead, they use attacks such as voice phishing and "pig butchering" fraud to build trust first. After several conversations, they made a harmful request. By using long interactions throughout the conversation, they hide the attack information, making it difficult for an automated scam detection system to detect.

We worked on COVA-X, a synthetic dataset of 10,985 multi-turn scam conversations. Then evaluates machine-learning models, which are XGBoost with TF-IDF, DistilBERT, and Longformer. The published XGBoost system relies on 28 engineered features that are not included in the paper, and so the full reproduction of the paper wasn't possible. We compared our recoverable TF-IDF component with a published paper and achieved 73.61% accuracy and a macro-F1 of 0.6835, compared to 78.43% accuracy and 0.7563 macro-F1. We measured the corpus and found that the published maximum token length of 874 is inconsistent with the measured median of 1,221 and maximum of 18,246. Which means DistilBERT truncates 94.5% of conversations and Longformer truncates 61.9%.

Our six-turn ablation analysis showed that restricting the model to early context reduced the F1 score for the "complied" class from 0.540 to 0.248, while the "rejected" class fell much less from 0.832 to 0.759. Typically, Compliance signals appear late in the conversation, making it difficult for real-time triage.

The study found six conversational indicators to serve as evidence for analysts. Among them, banking information requests and financial requests were retained with a masking-based faithfulness test. These indicators failed to reach the the 50% threshold with only 42.2% faithfulness, so they are presented as corroborating evidence rather than model explanations. We proposed a four-arm Human-AI triage framework. It adjusts the scam-type base-rate control and presents the calibrated outputs and corroborating evidence. We validated our Human-AI framework through statistical simulations instead of running a live trial with actual analysts. The results prove that classification accuracy alone cannot guarantee effective detection. Adding conversational context, confidence calibration, evidence quality, and human oversight makes the detection more accurate

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Chapter 1

Introduction

1.1 Background and Motivation

Modernizing world everyday brings us so many technical blessings along with the trouble no one expects. In recent time we are used to with social media and other platforms where we connect with many people for good undoubtedly via text but unfortunately some people wait for us to grab our attention and then control of our mind to manipulate us. Their intention is pretty clear, At the beginning they want to provoke the user by sentimental attachment or giving lavish offers and gradually they built the trust and make them ready to do whatever they want, at this point we think its alarming to let them do their unethical behaviour to harm any innocent.

Our traditional automated detection system catch this phishing or smishing but fails when the process is lengthy over several texts or continuous connection they collect the user data so professionally that at first it seems harmless but over the period of time the user become their digitally coerced victim .So, this project built the security system to protect these innocent users from the harmful activities they are not even assuming. We used COVA and COVA-X datasets to prevent this as there are social-engineering attacks shown as multi-turn conversations between language model agents. The expanded COVA-X dataset contains 10,985 synthetic conversations from eight different scam categories.

As its confidential to use any real data cause from the original chat history there are possibilities to leak someones privacy, private information about bank account and health or many other things to avoid these uncertainty we based our project on synthetic data consciously that relate with real wold crimes and also not exposing any ones identity. As machine learning learns the pattern gradually from the data generation procedure so sometimes it may feel unnatural conversation, artificial patterns, unrealistic relationship between variables therefore this project does not consider classification accuracy alone as enough evidence that a security system is successful.

1.2 Purpose and Goal of the Project

The intention of our project is to investigate is it possible to make more trustworthy and analyst oriented security support system of conversational social engineering detection. Our work contains five important goals.

Goal 1: Reproduce existing COVA-X baselines The first work we had done to check the existing result with our reproduced result we implemented the main models used in the COVA-X study. The models we had evaluate

- XGBoost with TF-IDF;
- DistilBERT;
- Longformer.

Reproduction is important important to ensure if the result depends on hidden details or not.

Goal 2: Evalute conversational context We checked the particular parts to ensure that, if the model performance do very or not depending on which section of the conversation is present in the model . The project therefore compares:

- complete conversations;
- the final 512 words;
- the first 512 words;
- the first six turns.

The purpose was not only to detect the particular part of the conversation which creating the most value on prediction but also to realize what possible prediction can the model make if the entire conversation is visible or the few texts can make prediction earlier. It become obvious if the entire conversation is present but observing few texts and detect scam is the actual task.

Goal 3: Accuracy prediction calibration The accuracy is an important factor cause a response of text or fraud alert depends on that. In case of higher accuracy than what the actually accuracy should be, the model can cause a serious inconvenience as a analyst can ignore a threat or a normal conversation can get affected by the flag, so the researchers want to adjust the models confidence after the model has been trained. So along with the correct prediction to maintain the trustworthiness we will use calibration techniques for example temperature scaling to reduce overconfidence.[1].

Goal 4: Provide analyst-facing evidence In this section our goal was to observe the model weather it just looks at common signs like creating urgency, asking for sensitive information, threatening or ask money from the user and make the decision cause these could be supportive evidence but not the final prediction and somethimes these words can be found as cue in regular conversation. So the project tests weather this indicators actually influenced the models prediction before using them as explanations.

Goal 5: Develop a Human–AI triage framework The final goal is to combine detection, confidence scores, supporting evidence, and human review into one workflow. The framework is designed to help security analysts make better decisions. Instead of making security decisions completely on its own, the system supports the analyst by providing useful information for human review.[2].

1.3 Organization of the Report

This report is organized into five chapters.

Chapter 1 introduces the research problem, motivation and objectives.

Chapter 2 reviews previous research in conversational social-engineering detection, confidence calibration, selective prediction and Human–AI security operations.

Chapter 3 describes the methodology, system architecture, dataset, software components, machine-learning models, calibration procedures and triage interface.

Chapter 4 presents the experimental investigation, classification results, temporal ablation, calibration analysis, evidence-indicator evaluation, impacts and critical discussion of methodological discrepancies.

Chapter 5 summarizes the findings, limitations and future research directions.

Chapter 2

Literature Review

2.1 Existing Research and Limitations

Research on social-engineering detection has advanced a lot recently, though current models still mainly focus on single suspicious messages, particularly those involving phishing and smishing. Machine-learning methods commonly depend on lexical features or traditional classifiers [3] for this task. This works well when a scam occurs in just one text. However, attackers are changing their behaviour and increasingly using longer conversations. Detecting an attack that develops slowly across several chats remains a major gap in current security systems.

A conversational attack usually starts by making the victim feel safe before suddenly introducing a false emergency. Only after a long chat might the attacker ask for money or sensitive credentials. Since the scam signals are distributed across several turns, standard detectors may not identify them easily. This is likely because they are designed to analyse one message at a time. Looking at the whole conversation instead may help overcome this weakness.

Changing how we examine these attacks also creates other issues. Processing long contexts and maintaining reproducibility are difficult, while confidence calibration and selective prediction remain challenging. Human-AI collaboration and explanation faithfulness are also difficult to achieve in practice. For instance, a detector might spot the scam after reading the entire conversation but still miss the early buildup in the first few messages. Similarly, a high-confidence prediction does not guarantee that the model is accurate.

2.2 Multi-Turn Conversational Detection

Recent Detectors are analysing long conversation instead of individual messages. These detectors combine multi-turn frameworks with an LLM-based agent to detect conversational social engineering [4, 5]. The COVA dataset introduced synthetic attacker and victim conversations [6], while COVA-X expanded this dataset to 10,985 conversations

across multiple scam categories [7].

The original COVA dataset performed well on TF-IDF-based XGBoost, but in COVA-X achieved 79.71% accuracy compared to XGBoost's 78.43% [6, 7]. So, the model's performance depends on the dataset's size and the amount of information it contains. However, we could not fully reproduce the reported results. XGBoost baseline has 28 engineered features, which are not provided in the COVA-X paper [7]. Though a lower score doesn't reflect that the paper was inaccurate. So, the result cannot be verified using available information, and so reproducibility is one of the research problems.

2.3 Long-Context Processing

Another problem in conversational social engineering is the length of the conversations. DistilBERT has a limit of 512 tokens, while the Longformer configuration used in this project has a limit of 1,024 tokens. Because of these limits, a large part of the conversations has to be truncated before it is given to the models. But in our reproduced corpus, the median conversation length is about 1,221 tokens, while the longest conversation has 18,246 tokens. Around 94.5% of the conversations are longer than 512 tokens, and 61.9% are longer than 1,024 tokens. This means that most of the conversations are too long to be fully processed by the models used in this project.

This can be a problem when the important information appears later in the conversation. For example, an attacker might first spend time gaining the victim's trust and later ask for money or other sensitive information. If the model only sees the beginning of the conversation, it is working with less information and may not recognise the actual threat in the later part of the conversation.

2.4 Confidence Calibration

Model accuracy and confidence are very important in the prediction. A classifier can make many predictions with different confidence levels, but that doesn't mean its confidence is reliable.

Guo et al. [1] showed that modern neural networks can have calibration problems. They introduced temperature scaling as a post-hoc method to improve calibration. Recent studies have highlighted that confidence calibration becomes even more critical in multiturn contexts and language models [8, 9].

In cybersecurity, Analysts may use confidence scores when deciding which alerts need more attention. For this reason, confidence needs to be calibrated using Expected Calibration Error (ECE), adaptive Expected Calibration Error (aECE), Maximum Calibration Error (MCE), Brier score, and Negative Log-Likelihood (NLL).

2.5 Human–AI Collaboration

In SOC environments , AI detectors are used as a decision support tool rather than a decision maker [10]. In this system, decisions from AI that have high confidence are handled directly, but uncertain decisions need human review or further analysis [11].

When there is a large number of alerts, the confidence-based decision is very useful. But if the confidence of the model is poorly calibrated, the whole system becomes unreliable, as there will be many false alerts [1].

Human analysts who depend on this automated alert from AI may face several problems. One such problem is automation bias, where an analyst accepts an incorrect prediction because the system gives a high confidence score [2]. When they trust the model too much and don't manually inspect alerts, many important information will be ignored. High confidence does not always mean that the prediction is correct.

Due to the high number of alerts, the SOC analyst faces a high workload. Subjective workload can be evaluated using the NASA Task Load Index (NASA-TLX) [12]

Our research is mainly focused on the reliability of models' confidence, and also provides supporting information so that analysts can make the final decisions [13]

2.6 Comparison of Existing Research

Table 2.1 compares the main research areas and their limitations.

Table 2.1: Comparison of Existing Research

Area	**Focus**	**Contribution**	**Limitation**
Traditional detection	Single messages	ML-based detection [3]	Limited conversational context
COVA / COVA-X	Multi-turn scams	Conversation datasets [6, 7]	Reproducibility and synthetic data
Long-context processing	Long conversations	More conversational context [4, 5]	Token limits may remove important information
Confidence calibration	Model confidence	More reliable confidence scores [1, 8, 9]	Less studied in long conversations
Human–AI collaboration	Analyst support	AI and human decisions [2, 10]	Risk of automation bias
Explanation and faithfulness	Supporting evidence	Helps explain predictions	Evidence may not reflect model behaviour

2.7 Research Gap

The literature review identifies four main gaps:

1. Research on Conversational social-engineering depends heavily on classification, while the reliability of model confidence gets less attention.

2. Applying confidence calibration to long conversations remains a major gap.

3. Early Detection in long conversations is much needed.

4. Detection, confidence calibration, evidence presentation, and Human–AI decision making are rarely evaluated together within a single framework.

We address these gaps with a framework which has conversational classification, context analysis, confidence calibration, corroborating evidence, and Human–AI triage altogether.

Chapter 3

Methodology

3.1 System Design

3.1.1 Overview

This project used a multi-stage methodology to analyse multi-turn social-engineering conversations. We processed the data with COVA-X v1.1 conversation corpus, then studied the text representation. We then did supervised classification, context manipulation, confidence calibration, evidence extraction, and Human–AI triage [6, 7].

The classifier models are separate from the confidence-calibration and analyst-support stages. We stored the model's prediction score and passed it to the later stages.

A complete conversation can contain information that only appears after the interaction has progressed. We identified restricted views of the conversation, including the first several turns and fixed-length head and tail representations. These different conversation views help us to understand the role of context in the model's predictions.[5, 7].

The system is divided into four layers. The first is the data layer. It prepares the conversational records and labels. Next, the detection layer generates predictions from the text. And then, the confidence and evidence layer. It processes model outputs and then adds information for the analyst. In the end, the Human–AI layer presents the results to a human reviewer.

The complete methodology is shown in Figure 3.1.

A model that gives a correct prediction may still have an unreliable confidence score. The probability it produces may not accurately show how likely the prediction is to be correct. Because of this, confidence calibration is treated as a separate stage, not as a part of model training [1].

3.1.2 Research Scope and Task Definition

The study uses outcome of the scam as its target. outcomes are grouped into three classes.

$$Y \in \{\text{Complied}, \text{Partial}, \text{Rejected}\}$$

Figure 3.1: Overall Methodology Framework

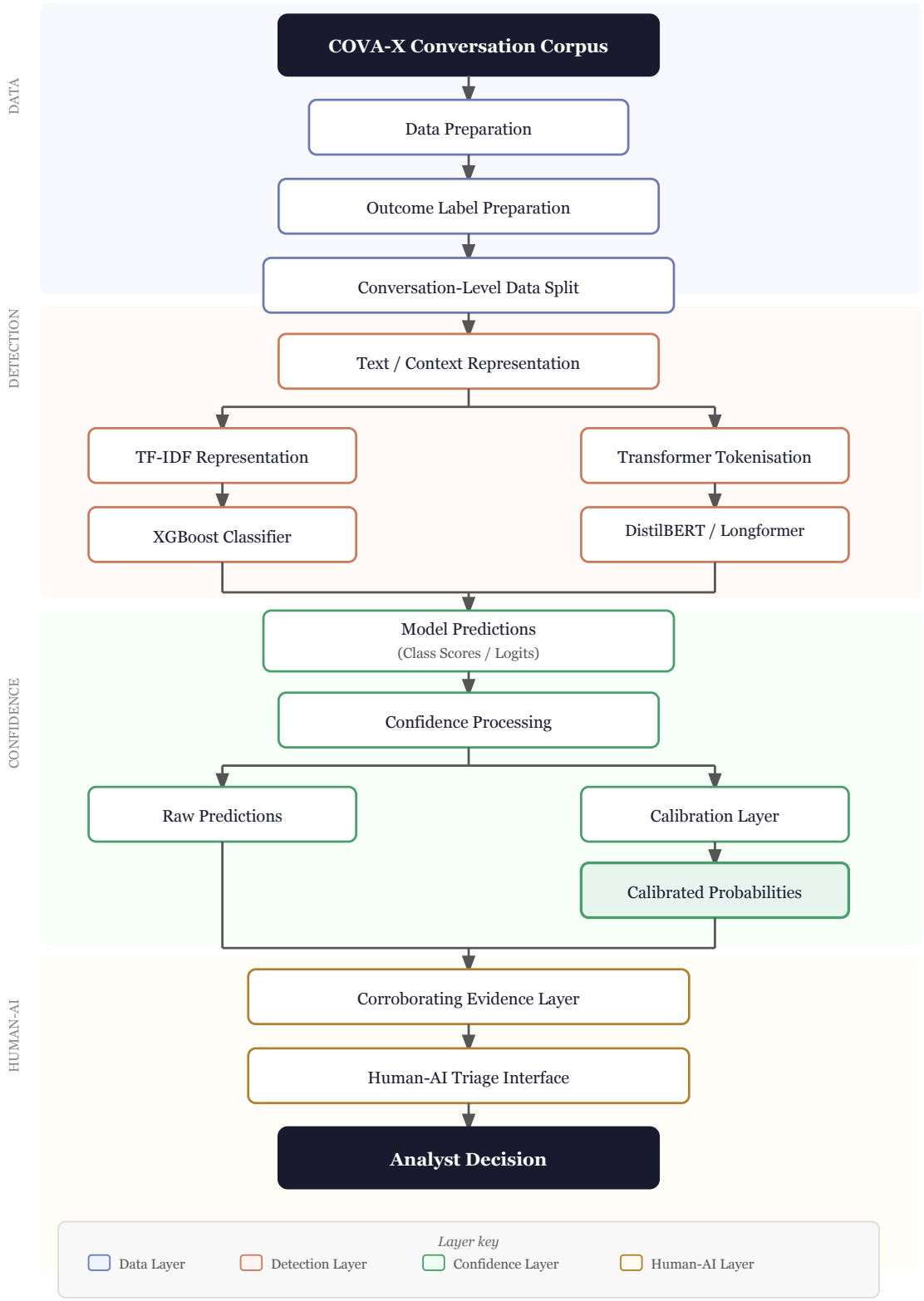


Figure 3.1: Overall Methodology Framework

The mapping is defined as follows:

- **Complied**: `successful_scam`
- **Partial**: `partial_compliance`
- **Rejected**: `scam_detected`, `quick_rejection`, and `verification_attempt`

The corpus distribution is approximately 15.2% Complied, 39.0% Partial, and 45.8% Rejected.

The classes are not evenly distributed. So, we used macro-averaged evaluation measures. The objective was to determine how conversational information can support scam outcomes. and to

3.1.3 Dataset and Partitioning

The COVA X v1.1 has been distributed across eight scam types. Each conversation has multiple turns and is labelled with outcome and metadata fields [6, 7]. The full conversation represents the sequence of informational exchanges between the attacker and the victim

The dataset was loaded and split into training, validation, and test sets (80/10/10). The split is performed at the conversation level, not at the message level. It has 8,788 training conversations, 1,098 validation conversations, and 1,099 test conversations. The partition was based on scam type and the three outcome classes. We used a fixed random seed of 42.

The validation set is used for model development decisions and calibration fitting, but the test set remains separated from the model, just used for final evaluation

3.1.4 Data Quality and Feature Restrictions

Before training, we checked the dataset for label consistency, conversation length, and possible overlap between partitions. Token lengths are measured using the `distilbert-base-uncased` tokenizer. The median length is 1,221 WordPiece tokens, with a 95th percentile of 2,638, a 99th percentile of 3,385, and a maximum of 18,246 tokens. We also checked the near duplicate conversations. These checks help separate dataset properties from model behaviour.

The classifiers use conversational text only. The following metadata and outcome fields are excluded from the detector:

- `scam_type`
- `victim_config`
- `attacker_config`
- `hostage_config`

- `legacy_contamination`
- `total_turns`
- `model_used`
- `outcome`

These fields are treated as metadata or control variables. The `outcome` field is excluded because it is the target, while `total_turns` describes the completed conversation and would not be available in the same form during early-stage prediction.

3.1.5 Text Construction and TF IDF Representation

For the pipeline, the whole conversations grouped into a single text representation while preserving turn order. For transformer models, the turn order remains the same before tokenisation. We applied the specified truncation strategy when the conversation exceeded the model limit.

TF IDF is used to represent each conversation numerically:

$$\text{TF IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

The vectoriser uses unigram and bigram features. It has a maximum vocabulary of 5,000, and sublinear term frequency weighting. It is fitted only on the training set. We applied it to the validation and test sets later. The sparse matrix is supplied to XGBoost.

3.1.6 XGBoost Classification

XGBoost was our main classical classifier. It has TF IDF representation. The model builds an ensemble of decision trees to reduce classification error.The objective can be written as:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k} \Omega(f_k)$$

where $l(\cdot)$ is the prediction loss and $\Omega(f_k)$ controls model complexity. Training set used for model training and for monitoring, we used the validation set. The validation macro F1 used for model development.

The used engineered features that are not publicly documented are not reproduced here. The implemented model represents the recoverable TF IDF component [7].

3.1.7 Transformer Classification

Two transformer models are implemented as three-class sequence classifiers:

- **DistilBERT**: `distilbert-base-uncased`

- **Longformer**: `allenai/longformer-base-4096`

DistilBERT uses a maximum of 512 WordPiece tokens and retains the final 512 tokens of longer conversations. We fine-tuned this model with AdamW and a learning rate of 2×10^{-5}, batch size 16, and up to 12 epochs. The best epoch is selected using the validation macro F1.

Longformer has a maximum of 1,024 tokens and uses the first 1,024 tokens of longer conversations. The learning rate is 2×10^{-5}, batch size 4, and up to 12 epochs. Mixed precision was applied during training when supported by the available GPU.

3.1.8 Prediction and Context Analysis

After training, each model produces scores for the three outcome classes. Transformer logits are converted to probabilities using:

$$P(y = k \mid x) = \frac{\exp(z_k)}{\sum_{j=1}^{K} \exp(z_j)}$$

where $K = 3$. The conversation identifier stored the probabilities.

Because conversations can exceed the available input limits, four text views are evaluated: the complete conversation, the final 512 tokens, the first 512 tokens, and the first six turns. The first six turns provide an early view of the interaction before the final outcome is observed. Comparing these views while keeping the model, labels, split, and vectoriser fixed allows the effect of visible context to be examined [5, 7].

3.2 Hardware and/or Software Components

3.2.1 Computational Environment and Software

The project is implemented in Python using Google Colab with a T4 GPU. Python is used for data processing, machine learning, statistical analysis, and visualisation. Pandas and NumPy are used for data handling and numerical operations. Scikit learn provides TF IDF vectorisation, data splitting, and evaluation functions.

3.2.2 Machine Learning Frameworks

XGBoost is used for the gradient boosted decision tree classifier.Then, Transformer models are implemented using PyTorch and Hugging Face Transformers. This allows comparison between the classical TF IDF pipeline and pretrained language models.

3.2.3 End to End Processing

The notebook follows the processing sequence shown in Figure 3.2.

Model predictions are stored after generation so that later calibration and evaluation can be performed without retraining the models.

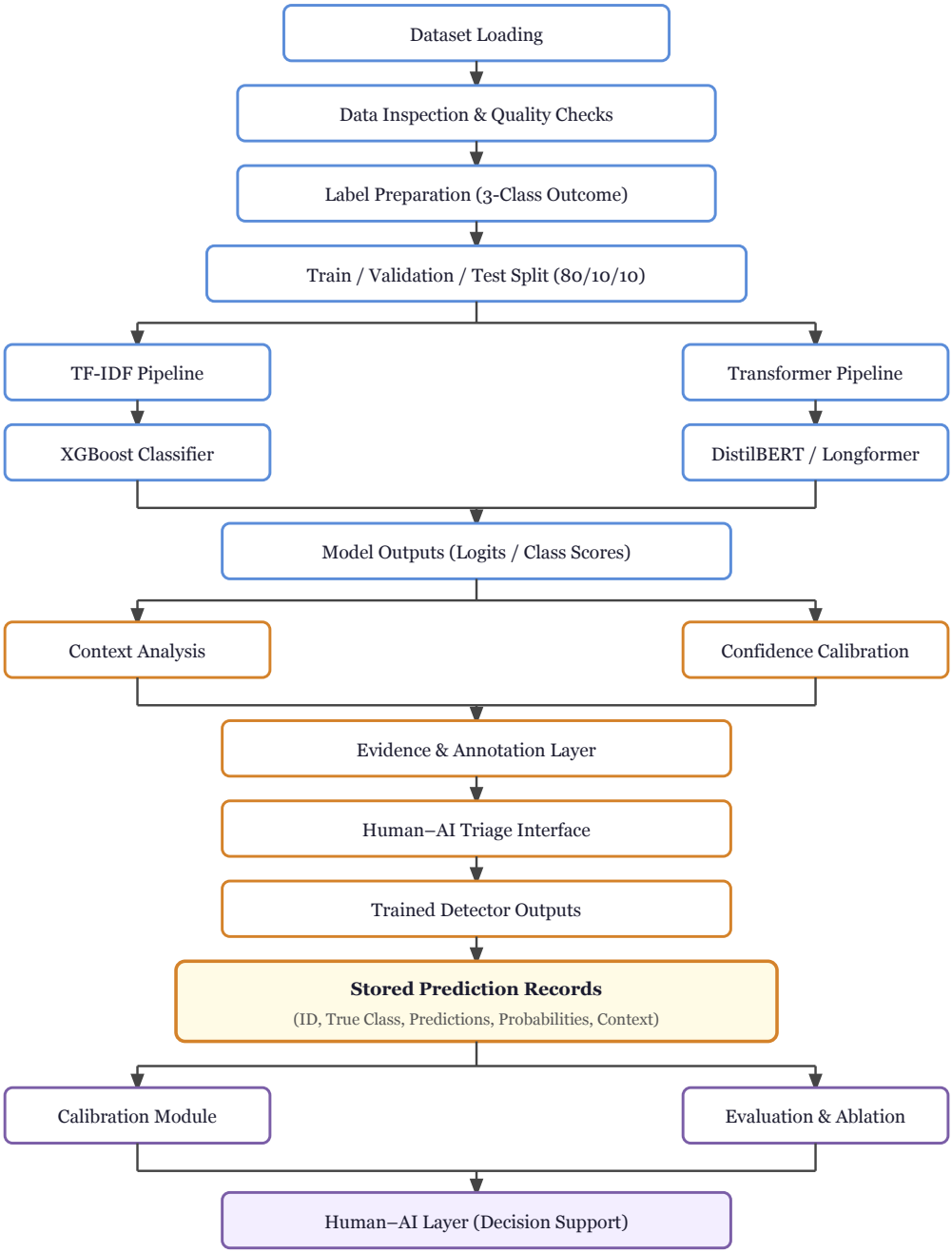


Figure 3.2: End-to-End Processing & Storage Architecture

Figure 3.2: End to End Processing and Storage Architecture

3.2.4 Data Loading and Label Preparation

The COVA X records are loaded into a dataframe, and the original outcomes are mapped to the three target classes shown in Table 3.1.

Table 3.1: Mapping of Original COVA X Outcomes to the Three Class Target

Original Outcome	**Target Class**
`successful_scam`	Complied
`partial_compliance`	Partial
`scam_detected`	Rejected
`quick_rejection`	Rejected
`verification_attempt`	Rejected

The mapping is applied before partitioning so that the same class definition is used throughout training and evaluation.

3.2.5 TF IDF and XGBoost Implementation

Each conversation is joined into a single text sequence. TF IDF converts the text into sparse features using unigrams and bigrams, a maximum of 5,000 features, and sublinear term frequency.

The vectoriser is fitted on the training set and applied to the validation and test sets. The resulting matrix is supplied to XGBoost, which outputs a predicted class and class probabilities.

3.2.6 DistilBERT Implementation

DistilBERT uses `distilbert-base-uncased` with a maximum sequence length of 512 tokens. When a conversation exceeds this limit, the final 512 tokens are retained.

The model is fine tuned using AdamW with a learning rate of 2×10^{-5}, batch size 16, and a maximum of 12 epochs. The best epoch is selected using validation macro F1, with random seed 42.

3.2.7 Longformer Implementation

Longformer uses `allenai/longformer-base-4096` with an input limit of 1,024 tokens. When the limit is exceeded, the first 1,024 tokens are retained.

The model uses AdamW with a learning rate of 2×10^{-5}, batch size 4, and a maximum of 12 epochs. The best epoch is selected using validation macro F1, with random seed 42. Mixed precision is used where supported by the GPU.

3.2.8 Context Ablation Implementation

Four text views are created from each conversation:

$$V_1 = \text{full conversation}$$
$$V_2 = \text{final 512 tokens}$$
$$V_3 = \text{first 512 tokens}$$
$$V_4 = \text{first six turns}$$

The same XGBoost pipeline, labels, split, vectoriser, and model settings are used across the four conditions. Only the visible conversational context is changed.

3.2.9 Confidence Calibration

Post hoc calibration is applied to the XGBoost prediction scores. Temperature scaling is fitted on validation logits by minimising negative log likelihood:

$$P_T(y = k \mid x) = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)}$$

The fitted temperature is then applied to the test predictions. Vector scaling is also evaluated as a multi class calibration method [1].

3.2.10 Calibration Metrics

Calibration is evaluated using ECE, adaptive ECE, MCE, NLL, and Brier score. Reliability diagrams are also used to compare predicted confidence with observed accuracy.

The theoretical Version 2 methodology additionally specifies signed calibration bias and Murphy decomposition. These are treated as proposed extensions because they are not part of the completed notebook analysis.

3.2.11 Faithfulness Test for Evidence Indicators

The retained indicators are evaluated separately from the classifier explanation. The selected cue text is isolated and the remaining conversation is masked. The classifier prediction is then checked to determine whether it remains unchanged.

The result is used as a faithfulness measure. The indicators are therefore presented as *corroborating evidence* and not as definitive explanations of the model's internal decision.

3.2.12 Leakage Control

For early prediction, only information available in the observed conversation prefix is used. Later turns that reveal the final outcome are excluded from the early prefix condition.

3.2.13 Model Output Storage and Reuse

For each prediction, the notebook stores the conversation identifier, true class, predicted class, class probabilities, model name, and context condition where available. These stored outputs are reused for evaluation and calibration, reducing repeated model training.

3.2.14 Human AI Triage Interface

The Human AI component presents the conversation, predicted class, confidence, relevant scam type information, and selected corroborating indicators to the analyst.

For the calibrated condition, calibrated probabilities are displayed instead of the raw model confidence. The analyst remains responsible for the final decision.

3.2.15 Human Evaluation Arms

The Human AI study design contains four conditions:

Table 3.2: Human AI Evaluation Conditions

Evaluation Arm	**Information Provided**
Human Only	Conversation only
Human + Raw AI	Conversation, AI prediction, and raw confidence
Human + Calibrated AI + Annotation	Conversation, AI prediction, calibrated confidence, and corroborating indicators
Human + Scam Type Base Rate	Conversation and known scam type base rate information

The main AI augmentation comparison is between the *Human + Calibrated AI + Annotation* condition and the *Human + Scam Type Base Rate* condition. This provides a control for information that may already be available from the scam category.

Chapter 4

Empirical Investigation, Results, Discussion, and Impacts

Chapter 3 described the methodology used to analyse multi turn conversational social engineering with the COVA X v1.1 corpus. This chapter presents the experiments carried out using the implemented notebook and discusses what the results mean for reproducibility, early triage, confidence reliability, supporting evidence, and Human AI decision support.

The results and discussion are presented together for each main experiment. This avoids repeating the same setup in separate sections and makes it easier to connect each result with its practical meaning.

4.1 Empirical Investigation and Findings

4.1.1 Corpus Verification and Data Integrity

The first step was to check whether the released COVA X v1.1 corpus matched the properties described in the original publications [6, 7]. Since the corpus contains only scam conversations, the classification task was based on the final outcome of each conversation instead of a scam versus legitimate classification.

The original outcome labels were grouped into three classes. Successful scams were mapped to *Complied*, partial compliance to *Partial*, and scam detected, quick rejection, and verification attempt to *Rejected*. The resulting class distribution is shown in Table 4.1.

Table 4.1: Outcome Class Distribution of the COVA X v1.1 Corpus

Collapsed Class	**Original Terminal State(s)**	**Corpus Prevalence**
Complied	`successful_scam`	15.2%
Partial	`partial_compliance`	39.0%
Rejected	`scam_detected`, `quick_rejection`, `verification_attempt`	45.8%

Several other properties of the corpus were checked. First, the published description states that conversations shorter than ten turns were excluded. However, 631 such conversations are present in the released corpus, which is 5.7% of the dataset. The shortest conversation contains two turns. There is also a strong concentration at the 21 turn limit, with around 35.1% of conversations ending there. The high number of verification attempt cases at this point suggests that the generation limit may have affected how some conversations ended.

The second check concerned the reported legacy contamination. The published material refers to 1,060 conversations with this condition, but the same flag was not present in the inspected released records. A five fold logistic regression using conversation prefix text produced an AUROC of 0.9999 for separating the legacy condition from the non legacy condition within the government impersonation subset. This shows that the condition can be recovered very strongly from the conversation text.

The third check focused on token length. The publication reports a maximum of 874 tokens, but the released corpus is much longer. Using the `distilbert-base-uncased` tokenizer, the median length was 1,221 tokens, the 95th percentile was 2,638, the 99th percentile was 3,385, and the maximum was 18,246 tokens. As a result, the 512 token DistilBERT limit truncates 94.5% of the conversations, while the 1,024 token Longformer limit truncates 61.9%.

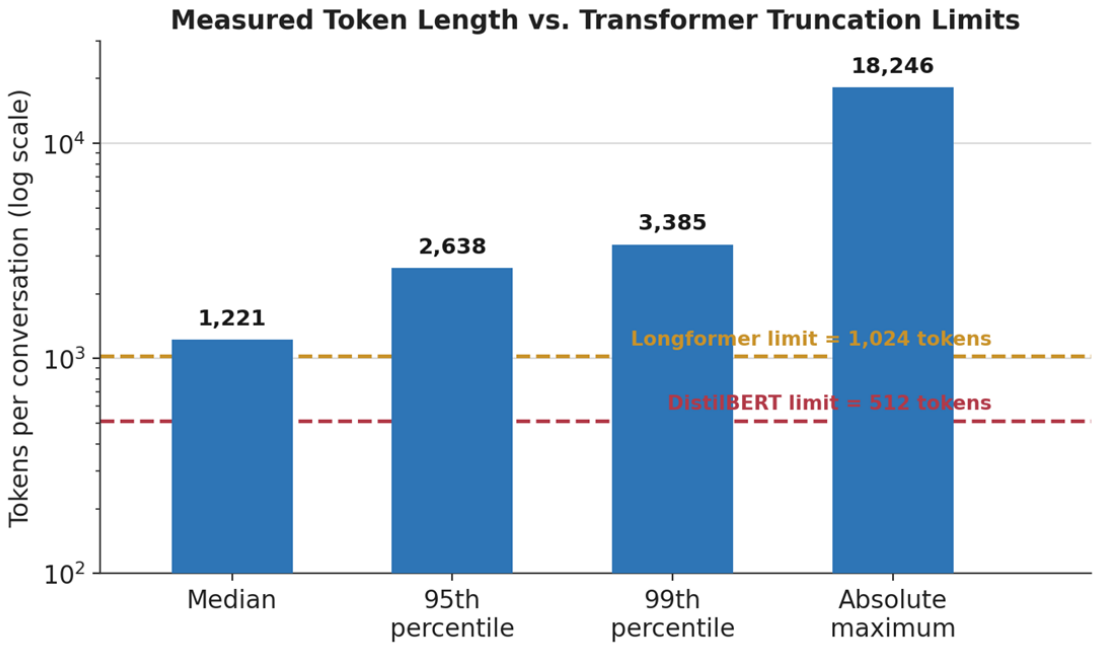


Figure 4.1: Measured per conversation token length percentiles using the `distilbert-base-uncased` tokenizer compared with the 512 token DistilBERT and 1,024 token Longformer input limits.

These checks matter because the dataset itself affects how the model results should be understood. A model cannot be treated as using full conversational context when most

conversations are longer than its input limit. The corpus audit is therefore part of the evaluation, not just a separate data cleaning step.

4.1.2 Reference Baselines and Configuration Mediation

The next experiment checked how much of the predictive signal could come from the way the synthetic conversations were generated. Three reference points were used on the same six turn horizon: a constant class base rate, a scam type base rate, and the full text model.

The constant base rate gave an AUROC of 0.500. The scam type base rate reached 0.7192, while the full text model reached 0.8338. The full text model therefore gave an improvement of 0.1180 AUROC over the scam type baseline.

The configuration mediation test also showed that several simulated scenario variables could be recovered from the conversation prefix. Scam type was recovered with 99.74% accuracy, simulated victim awareness with 89.08%, and trust related configuration with 88.79%. A separate harm model using these predicted configuration variables achieved an AUROC of 0.7780.

These results show that information about the generation setup is present in the conversation itself. This does not prove that the text classifier directly uses the metadata, but it does show that some of the scenario configuration can be inferred from the text. This is important when interpreting performance on a synthetic dataset.

4.1.3 XGBoost Reproduction and Performance Gap

The main classical baseline was reproduced using the recoverable TF IDF representation described in Chapter 3. The vectoriser used unigram and bigram features, a maximum vocabulary of 5,000, and sublinear term frequency. Three XGBoost settings were compared: the default configuration, balanced sample weights, and the tuned configuration used for the main reproduction.

Table 4.2: XGBoost Baseline Reproduction Compared with the Published COVA X Reference

Model Configuration	Accuracy	Macro F1	Weighted F1	F1 Complied / Partial / Rejected
Published reference with 28 undisclosed features	0.7843	0.7563	0.7825	0.677 / 0.728 / 0.864
Notebook best, XGBoost + TF IDF	0.7361	0.6835	0.7280	0.540 / 0.678 / 0.832
XGBoost defaults	0.7152	0.6626	0.7086	0.514 / 0.659 / 0.815
XGBoost with balanced weights	0.7207	0.6863	0.7170	0.597 / 0.639 / 0.823

Published figures are reproduced from the original COVA X publications [6, 7]. The 28 engineered features are not sufficiently documented for exact reconstruction.

The best reproduced XGBoost model achieved 73.61% accuracy and a macro F1 of 0.6835. This is about 4.8 percentage points below the published accuracy and 0.073 below the published macro F1.

Most of the difference appears in the complied class. The reproduced model correctly identifies 77 of 165 complied cases, compared with 108 in the published result. The rejected

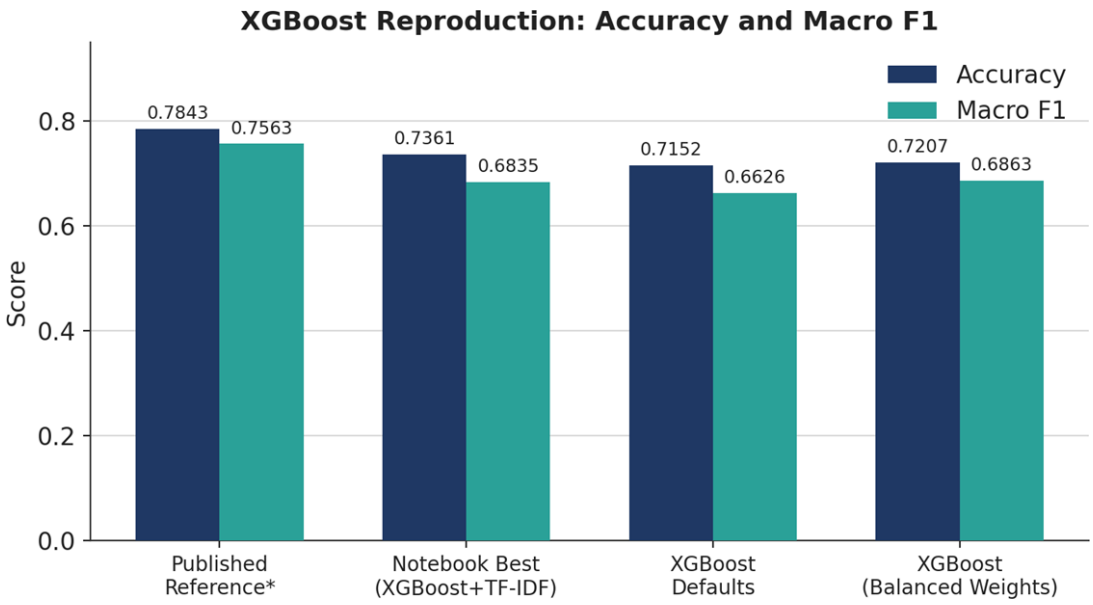


Figure 4.2: Accuracy and macro F1 for the published reference and the reproduced XGBoost configurations.

class is much closer, with 451 correct predictions in the reproduction compared with 446 in the published result.

This does not mean that the published result is wrong. The main issue is that the original system used 28 engineered features that are not documented well enough to reproduce. Those features may have helped the model distinguish different levels of victim compliance, but the current experiment cannot show how much of the performance gap they explain.

4.1.4 Context Ablation and the Early Triage Problem

The context ablation experiment examined what happens when the amount and position of visible conversation text are changed. The same labels, split, vectoriser, and model settings were used in all conditions. Four views were tested: the full conversation, the final 512 tokens, the first 512 tokens, and the first six turns.

Table 4.3: Context Ablation Results Across the Four Conversation Views

Context View	Accuracy	Macro F1	F1 Complied / Partial / Rejected
Full conversation	0.7361	0.6835	0.540 / 0.678 / 0.832
Final 512 tokens	0.7161	0.6575	0.491 / 0.677 / 0.805
First 512 tokens	0.6833	0.6268	0.466 / 0.630 / 0.784
First six turns	0.6515	0.5447	0.248 / 0.628 / 0.759

The same labels, split, vectoriser, and model settings were used across all conditions.

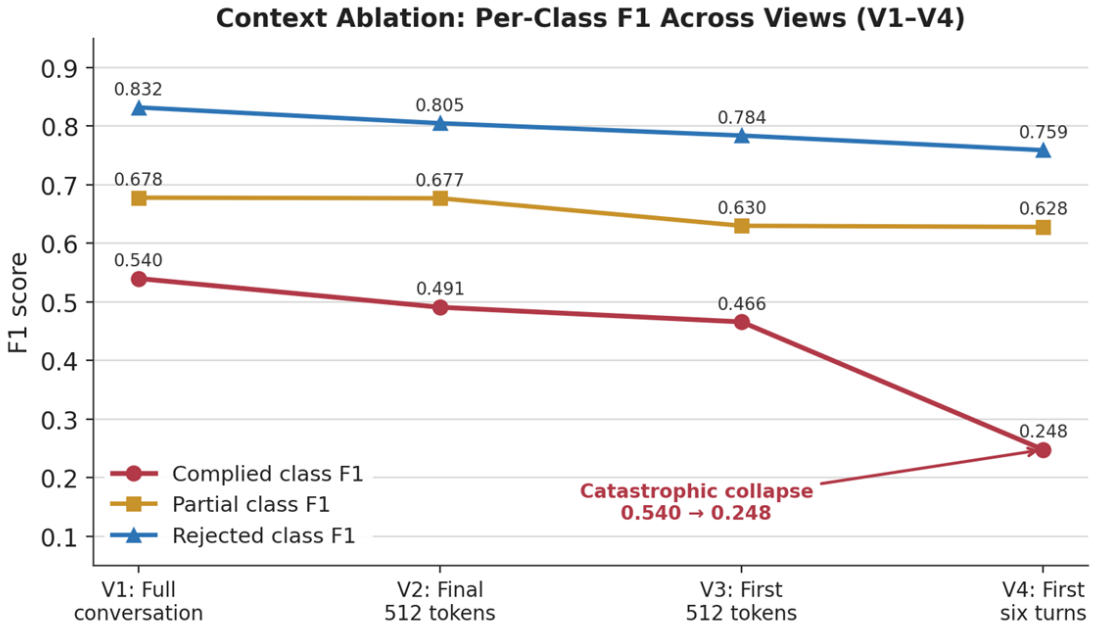


Figure 4.3: Per class F1 across the four context views, showing the reduction in complied class performance as the visible context is restricted to the early conversation.

At the same 512 token limit, the final 512 tokens performed better than the first 512. Accuracy increased from 0.6833 to 0.7161 and macro F1 increased from 0.6268 to 0.6575.

The largest drop occurred with the first six turns. Complied F1 fell from 0.540 with the full conversation to 0.248, while rejected F1 fell from 0.832 to 0.759.

This shows that information related to eventual compliance is more visible in the later part of the conversation. The full conversation model is therefore better suited to retrospective classification than to early triage. In a real setting, this matters because the system needs to provide useful information before the harmful behaviour has already happened.

The result also shows why early and full conversation views need to be kept separate. The early prefix does not include later turns that reveal the final outcome, while the full and tail views can contain information that would only become available later.

4.1.5 Transformer Fine Tuning and Context Constraints

Two transformer architectures were fine tuned using the configurations described in Chapter 3. DistilBERT used a maximum of 512 tokens and retained the final part of longer conversations. Longformer used 1,024 tokens and retained the first part.

Longformer achieved a macro F1 of 0.6584, compared with 0.6320 for DistilBERT. This follows the same ranking reported in the published results.

However, Longformer does not actually see the whole conversation. The corpus audit showed that 61.9% of conversations are longer than its 1,024 token limit.

The two models also use different token budgets and different truncation directions.

Table 4.4: Transformer Fine Tuning Results Compared with Published Figures

Architecture	Notebook Accuracy	Notebook Macro F1	Published Accuracy	Published Macro F1
DistilBERT, tail 512	0.6815	0.6320	0.7753	0.7632
Longformer, head 1,024	0.6915	0.6584	0.7971	0.7786

Published figures are reproduced from the original COVA X publications [6, 7].

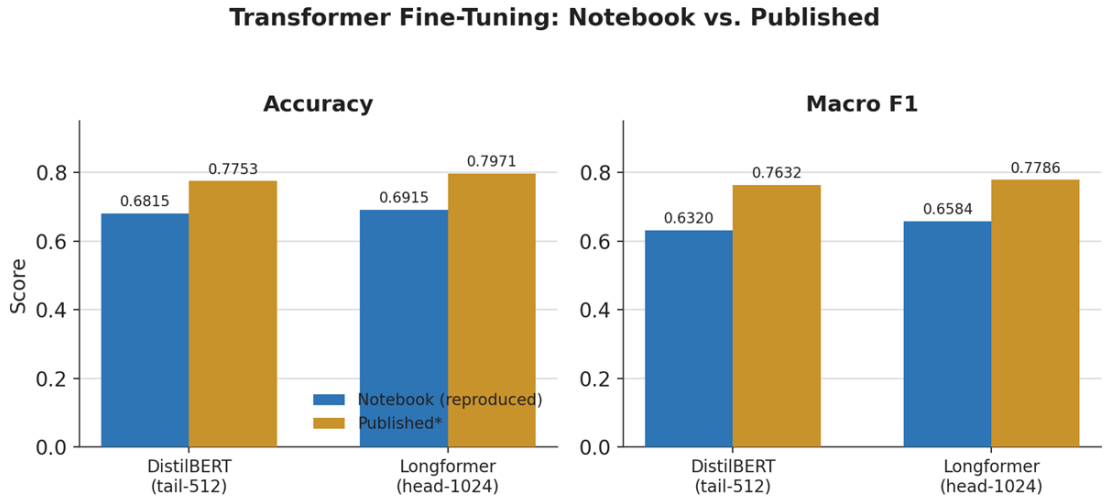


Figure 4.4: Notebook reproduced accuracy and macro F1 compared with the published figures for the evaluated transformer architectures.

Longformer keeps the first 1,024 tokens, while DistilBERT keeps the last 512 tokens. Because of this, the experiment does not isolate architecture from context availability.

There is also a gap between the theoretical Version 2 methodology and the executed notebook. Version 2 proposed DeBERTa v3 base and a logistic regression baseline, but those components were not implemented. Longformer was used instead. The transformer results should therefore be read as results from the executed configurations, not as a complete implementation of the theoretical Version 2 plan.

4.1.6 Confidence Calibration and Probability Reliability

A model's confidence values cannot be depended on accuracy only. To check this separately, the XGBoost prediction scores and values were calibrated after training. It has been calibrated by using both Temperature Scaling and Vector Scaling — each fitted on validation predictions first, and it has then been applied to the test set.

Table 4.5: Post Hoc Calibration Results for the XGBoost Classifier

Calibration Method	ECE	aECE	MCE	NLL	Brier	Accuracy
Uncalibrated	0.0440	0.0433	0.6090	0.6037	0.3615	0.7361
Temperature Scaling	0.0322	0.0281	0.2356	0.5990	0.3587	0.7361
Vector Scaling	0.0278	0.0405	0.1238	0.5963	0.3581	0.7325

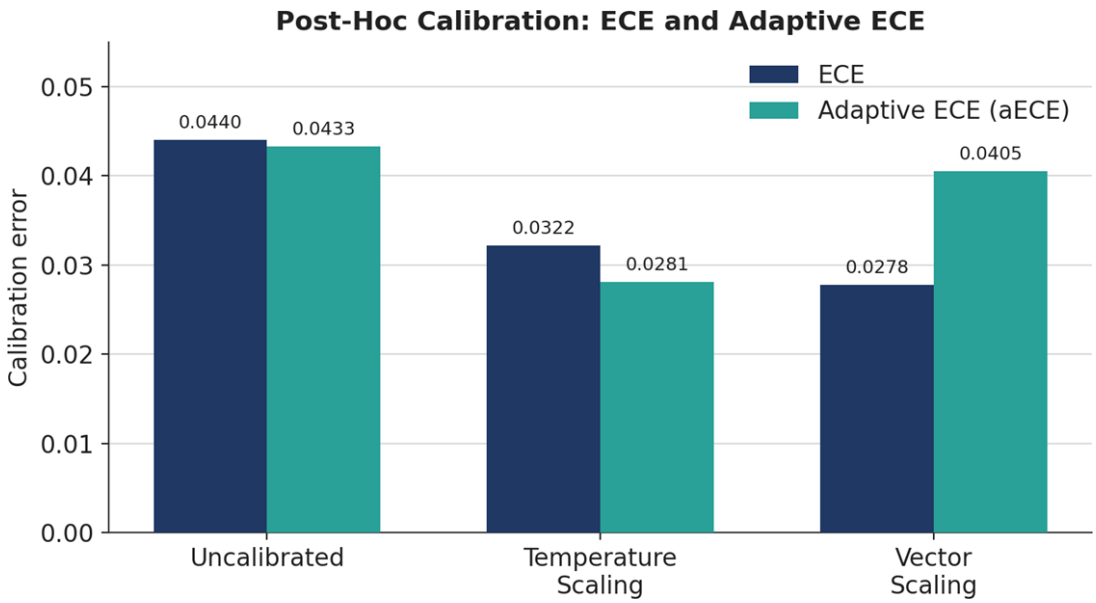


Figure 4.5: Expected Calibration Error and adaptive Expected Calibration Error for the uncalibrated and calibrated XGBoost predictions.

Temperature Scaling settled on $T = 1.2082$. Since this is above one, it indicates the raw, uncalibrated probabilities were somewhat overconfident to start off. If we can apply the correction ECE comes down from 0.0440 to 0.0322 and aECE from 0.0433 to 0.0281, with accuracy left untouched at 0.7361. Vector Scaling continued to push ECE even lower, to 0.0278, and finished bottom on MCE, NLL and Brier score on all three measures. The improvement was not without a cost, though, as accuracy slightly fell from 0.7361 to 0.7325. There is a true compromise between how well the probabilities are calibrated and whether there are any changes to the original classification decisions. After Temperature Scaling, the overall Brier score dropped from 0.3615 (uncalibrated) to 0.3587; after Vector Scaling, it went down further to 0.3581. It is a minor but real advance. It doesn't, sadly, provide that information either — the Version 2 methodology involved a Murphy decomposition of the Brier score and a signed calibration-bias measure: both those items were not included in the executed notebook. The overall numbers are better; how this improvement is distributed between the reliability, resolution and uncertainty is still not known.. Where a system is used by analysts, Temperature Scaling is the more conservative approach, as it does not modify the underlying classification boundary in any way. Vector Scaling yields a slightly better performance on some of the probability quality measures, while sacrificing a little on classification accuracy.

4.1.7 Corroborating Evidence and Faithfulness

A flagged conversation would be a place where a short highlighting of key indicators of an important part of the conversation would be helpful to an analyst in their review of the conversation. However, it is not the same thing to say that a cue that an analyst

uses to explain what a model is doing internally is useful, and that it is an explanation. The 6 candidate indicators underwent rule-based text matching: one-time password requests, banking information requests, suspicious URLs, urgency language, general financial requests, and password or credential requests. They were all examined with regard to frequency and association with the result. The analyst-facing layer only had banking information requests and financial requests pass this screening.

Table 4.6: Corroborating Evidence Indicator Screening Outcome

Candidate Indicator	**Screening Outcome**
One time password request	Excluded — insufficient coverage or weak association
Password credential request	Excluded — insufficient coverage or weak association
Banking information request	Retained as an analyst-facing indicator
Suspicious URL	Excluded — insufficient coverage or weak association
Urgency language	Excluded — insufficient coverage or weak association
Financial request	Retained as an analyst facing indicator

The two of these were then subjected to the masking technique as described in Chapter 3, which involves removing the presented text from the input and presenting only the masked text to the classifier and observing what it does. The result recovered approximately 42.2% of the above chance discrimination which was less than the threshold of 50% that was set prior to the test for calling a result model evidence or just a correlate. For this reason, these indicators are used as evidence, rather than definitive explanations of the behavior of the model. A cue's utility to an analyst is not equivalent to a cue's effect in causing a classifier's decision. Furthermore, mixing the two would be an exaggeration of what has been demonstrated. This is actually a restriction on what the annotation layer can assert, and this is not necessarily an indication that the annotation component has failed.

4.1.8 Speaker Channel Analysis

One part of the investigation focused on the origin of the predictive signal – the attacker's message, the victim's reply, or the entire interaction. Unidirectional classification was re-done for the victim-only, attacker-only, and combined speaker representation. Unfortunately, the execution artefacts provided for this thesis did not contain a complete numerical breakdown of accuracy and F1 in all three conditions. Because of this disparity, the experiment is presented below not as evidence that one of the two speaker channels performs better than the other, but as a qualitative investigation of the differences in the two channels. The absence of data in this space is significant for interpretation. Financial

requests and financial language can appear on either side of a conversation: the person requesting the financial information, or the person's who is hesitant, seeking confirmation or offering to cooperate. If there isn't a complete quantitative comparison, it's difficult to determine if it's mostly due to the tactics of the attackers, the reactions of the victim or both.

4.1.9 Human AI Teaming Simulation

The Human-AI triage design was built around four major conditions: Human Only, Human plus Raw AI, Human plus Calibrated AI plus Annotation, and Human plus Scam-Type Base Rate. The last of these acts as a control, accounting for information an analyst might already have simply from knowing the scam category.

Table 4.7: Human AI Evaluation Conditions Used in the Simulation

Arm	**Condition**	**Purpose**
1	Human Only	Establishes the human decision baseline
2	Human plus Raw AI	Measures performance with uncalibrated AI assistance
3	Human plus Calibrated AI plus Annotation	Evaluates the proposed calibrated AI support condition
4	Human plus Scam Type Base Rate	Controls for information that may be available from scam category alone

There was no actual involvement of participants at this stage. Rather, synthetic participant records were created using a Monte Carlo simulation to assess correct behaviour of the statistical analysis pipeline itself. Decision times were normally distributed around lognormal distributions. Furthermore, the confidence values normally distributed about latent per-condition skill parameters from 0.62 to 0.74. The Shapiro–Wilk normality test was used to determine if pooled residuals were normally distributed and was found to be $W = 0.908$, $p = 0.1991$, so the pipeline was continued to a one-way ANOVA. This returned $F = 14.267$, $p = 0.0014$ which was followed by Holm-corrected pairwise comparisons against the Arm 4 base-rate control. The simulated Human Only AUROC falls within the designed operating range of 0.60–0.95 — this is the point of the exercise, to make sure that the statistical pipeline behaves as designed and passes the simulation level readiness gate. What it does not do, and cannot do, is show that calibrated AI assistance would actually help real security analysts. That is a separate empirical question, and answering it properly would require a genuine participant study, not a simulation.

4.1.10 Comparative Positioning Against the Literature

The models produced are audited models for future research on conversational social engineering to be compared with, which should be handled with care. The cited studies

differ from this project with respect to dataset, objective, modality, architecture, and/or evaluation procedure, as a combination of two or more in some cases.

Table 4.8: Comparative Benchmark of Reported Detection Performance

Model or Framework	Primary Detection Approach	Peak Accuracy	Peak F1
COVA X notebook, XGBoost + TF IDF	Static lexical retrospective classification	0.7361	0.6835
COVA X notebook, Longformer 1,024	Truncated contextual encoding	0.6915	0.6584
ConvoSentinel style approach	Retrieval augmented generation and conversation analysis	0.9800	0.9721
SE OmniGuard style approach	Dynamic psychological state tracking	0.8130	0.8150
Multimodal vishing defence	Text and audio fusion	N/A	0.9950

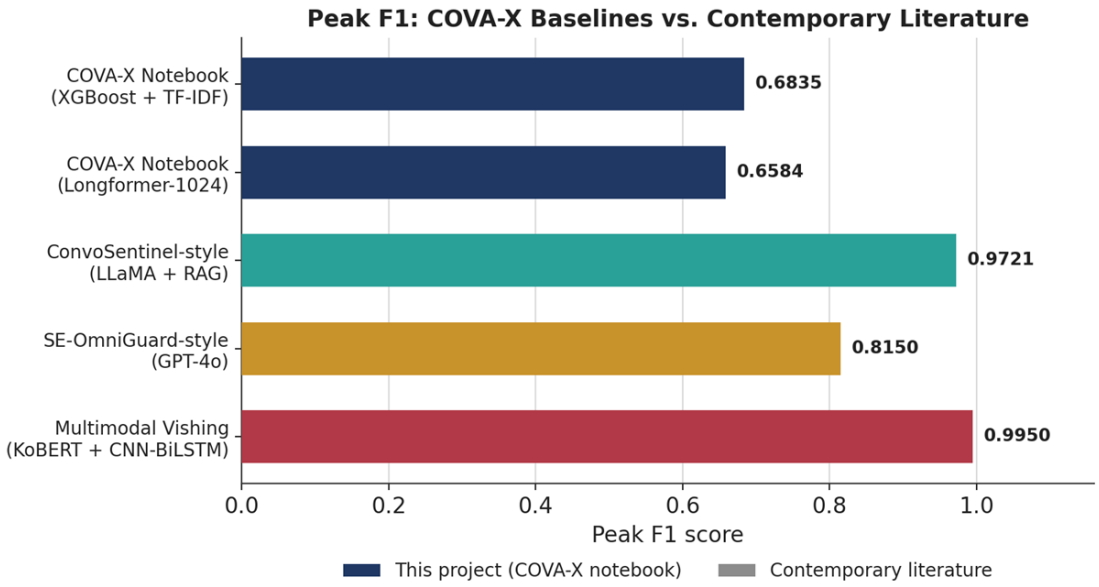


Figure 4.6: Reported peak F1 values for the reproduced COVA X baselines and selected contemporary conversational social engineering approaches.

The headline numbers for the contemporary systems cited here are significantly better, especially for systems leveraging retrieval-augmented generation, dynamic psychological tracking, or multimodal processing, with macro F1 scores 0.9721, 0.8150, and 0.9950, respectively, compared to 0.6835 and 0.6584 for the reproduced XGBoost baseline and the reproduced Longformer, respectively. Obviously, none of this is a simple ranking as the experimental conditions are not identical. The comparison is useful in context, that is, as an audited research reference point, not as a system that is ready to be deployed as an actual operational system.

4.2 Integrated Discussion

4.2.1 Interpretation of the Corpus Findings

Nearly every experiment in this chapter needs to be read in light of the corpus verification findings discussed earlier. Conversations that should have been filtered out slipped

through, turn counts cap off sharply at 21, and the token-length distribution runs well beyond what the original publication reported. Together these irregularities point to a dataset whose shape reflects the way it was generated at least as much as the behaviour it was meant to capture. That doesn't make the corpus unusable. It does mean, however, that every performance figure in this chapter should be treated as a product of those generation conditions rather than a clean, isolated measurement of how people actually talk in this setting. The configuration mediation results back this up. Scam type and other scenario details can be pulled out of the conversation prefix with surprisingly high accuracy, suggesting that a meaningful chunk of the model's predictive power comes from structural leftovers of the generation process rather than from anything that resembles genuine social-engineering behaviour. That finding alone justifies excluding scenario metadata from the detector's feature space. It's also the reason the scam-type base-rate condition was built into the Human-AI evaluation to begin with. Without it, whatever benefit the detector appears to offer could just as easily be the model rediscovering the scenario label rather than learning anything real about compliance.

4.2.2 Reproducibility and the XGBoost Performance Gap

If there's one result in this project that makes the case for reproducibility as a discipline, it's the XGBoost experiment. Landing at 73.61% accuracy against a published 78.43% is a real gap, and there's no clean way to pin it entirely on the missing engineered features, since the published documentation doesn't go into enough detail to isolate that effect. The class-by-class breakdown tells a more useful story than the headline number. The rejected class comes close to matching the published figure, while the compliance class is where most of the gap opens up. One plausible explanation is that the original 28 engineered features carried information that was particularly good at distinguishing degrees of victim compliance. That has to stay a guess, though, since those feature definitions were never made public. It's also worth flagging separately that the proposed 60/15/25 train-validation-test split was never actually used. The notebook ran on an 80/10/10 split instead. The results above are valid for that implemented version of the experiment, but they can't tell us anything about how performance would have looked under the originally planned partition.

4.2.3 Context Constraints and Retrospective Detection

Of everything in this chapter, the temporal ablation probably has the most direct practical implication. Removing later conversation turns hits the compliance class hard, while the rejected class barely notices the missing context. In other words, the signal needed to catch eventual compliance seems to show up late, right around the point where compliance is already happening. As things stand, the detector works better as a way of confirming what already happened than as a way of predicting what's about to happen. That's not a distinction to brush past. A system that only recognises harm after the victim has

already given away the relevant behaviour is solving a different problem than one that has to decide, in the middle of a conversation, whether to step in.

4.2.4 Transformer Performance and Context Limitations

Longformer beating DistilBERT in the reproduction does at least match the direction of the published comparison. But the geometry of the corpus — most conversations run well past even Longformer's window — makes it hard to read this as support for giving the model unrestricted full context. Two things are changing between the models at the same time: how many tokens each one gets, and which end of the conversation those tokens come from. Longformer looks at the first 1,024 tokens; DistilBERT looks at the last 512. Because both variables move together, there's no way in this experiment to separate the effect of architecture from the effect of context availability. The transformer results are best understood, then, as a faithful reproduction of what was actually implemented, not as a controlled test of how context length interacts with model architecture

4.2.5 Confidence Reliability and Appropriate AI Use

Accuracy on its own clearly isn't enough when a model's output is meant to support a human decision-maker, and the calibration results make that point concrete. Temperature Scaling improved both ECE and aECE without touching accuracy at all, while Vector Scaling pushed further on several probability-quality metrics but gave up a small amount of accuracy to get there. That's a reasonable case for building calibration into any triage-support system. What it can't settle, because the Murphy decomposition wasn't carried out, is exactly where the overall Brier-score improvement is coming from — whether reliability, resolution, or uncertainty is doing most of the work. Temperature Scaling still comes out as the more conservative option for this implementation, since it sharpens confidence quality while leaving the original classification decisions exactly as they were.

4.2.6 Evidence Presentation and Faithfulness

The evidence experiment argues for similar caution on the explanation side. Indicators that work well for shorter, message-based detection don't automatically carry over to long conversational social engineering, and the faithfulness score here is what backs that up. Banking-information and financial-request indicators are still worth putting in front of an analyst, but the recovery level measured here falls short of the threshold set in advance for calling them genuine model evidence. The interface presents them, appropriately, as corroborating evidence rather than anything stronger. If anything, that distinction matters more in a busy security operations environment, not less. An analyst working under time pressure shouldn't assume a highlighted phrase is proof that the classifier's decision actually hinged on it.

4.2.7 Human AI Collaboration and Automation Bias

What the Human-AI simulation actually confirms is narrower than it might first sound: the statistical pipeline runs its intended decision process correctly. It doesn't tell us whether AI assistance would help a real analyst in practice. The scam-type base-rate control is probably the single most important design choice in this whole framework. Given how much predictive value the configuration mediation experiment found sitting in scenario information alone, a straightforward Human Only versus Human-AI comparison risks making the detector look more useful than it is. Comparing calibrated AI assistance against the base-rate control instead gives a cleaner picture of what the AI is actually adding. So this simulation should be treated as what it is — a methodological validation step — and nothing more. It is not a study of human performance.

4.2.8 Methodology and Execution Gaps

Three gaps separate the theoretical Version 2 methodology from what actually got run in the notebook.

1. **Data partitioning:** the theoretical design calls for a 60/15/25 train/validation/test split; the executed notebook uses 80/10/10 instead.

2. **Model selection:** DeBERTa v3 base and a logistic regression baseline were both specified in the theoretical methodology, but neither one shows up in the executed notebook, which trains Longformer instead.

3. **Calibration reporting:** a Murphy decomposition and a signed calibration-bias measure were planned but never implemented; the notebook reports only the aggregate Brier score, alongside the other calibration metrics discussed above.

None of this invalidates the experiments that were actually carried out. It does mean that any claims about the complete Version 2 methodology need to be scaled back to match what was really done.

4.2.9 Overall Research Contribution

Taken together, the experiments in this chapter add up to a fairly connected story. The published COVA-X baseline can't be reproduced exactly from what's publicly available, and the released corpus carries structural quirks that complicate any straightforward reading of model performance. The strongest signal for eventual compliance sits late in the conversation, which means the current system works better as a retrospective tool than as an early-warning predictor. Post-hoc calibration can sharpen confidence quality without changing the underlying classification decisions. Observable conversational indicators are worth keeping in the analyst-facing layer, but they shouldn't be presented as explanations unless they clear a stronger faithfulness bar than they do here. And the Human-AI framework offers a structured basis for evaluation going forward, even if the current simulation

falls well short of proving any real benefit to actual analysts. None of this adds up to a finished, autonomous security system — and it was never meant to. What this project offers instead is an empirically grounded framework: one that lays out the conditions under which conversational detection, confidence calibration, corroborating evidence, and human review might reasonably work together.

4.3 Impacts and Design Constraints

The strong performance of the AI models is not only the subject of this Study. The result of this study will be used by people in the different institutions, security operations, privacy, organizational culture and computing resources, which will have a sustainable impact in the future. There is a important thing is this study is conducted by Artificial conversation and depend on simulated AI evaluation. Hence the possible impacts are discussed about those are not the real proven results rather how these impacts in future in the place of engineering field or the real implementation and we have to consider it for future.

4.3.1 Societal, Health, and Psychological Impact

Real life can be hampered many ways because of the social engineering, like financial loss, mental anguish, disgrace, Humiliation, Erosion of trust. If any system can find out high-risk conversation earlier, In that case theoretically, security team or human analyst can take steps ealier. As a result, the limited security resources can used towards the high-risk incidents more effectively. However, there is no proof has been found in this study that this system reduce the victim of social engineering or the reduce the Psychological distress. The COVA-X corpus is an artificial and there is no real affected person were involved in this study. if there is a possible benefit remain here that will be indirect and that benefit will be get in the real life it will be depend on two thing One is whether the system will be used in the future successfully or human analyst can work on its flags.

4.3.2 Safety and Human AI Reliance

If security system gives wrong decision then there will be problem from both sides. False Negative, If System can failed to indicate harmful or fraud interaction then the harmful interaction will go on without any hindrance. False Negative: If the system defective count the legitimate thing as risky then human analyst should give unnecessary time into it. Human-AI architecture does not take any security decision. Always a human analyst will take the decision. Confidence calibration is a thing that make numerical confidence signal that truly refers something but corroborating-evidence layer will give lots of information without crossing the line with its own description. The objective of this system is helping human judgmental ability not the taking place of human wisdom.

4.3.3 Organisational and Security Impact

Giant security team always gets the so many alerts that they can not check the problem equality properly. This type of problem exactly can be solved by AI-assisted triage system. Those case are most importantly checked the Ai can indicate the AI analyst's attention here. An inaccurate detector can add more workload rather then reduce it. If Ai cannot be calibrated properly, if it gives false positive, or gives such a evidence that mislead the analyst then the mental workload and cognitive burden increased. Therefore this proposed workflow treats not only AI is accurate, how much time it takes to take a decision, how much analyst feels confidence, how much analyst has to take workload and how much analyst depend on Ai , all of these come in the design considerations.

4.3.4 Legal and Privacy Considerations

The data of the real cybersecurity conversation can be generally human name, numbers, financial information, and it can includes different types of personal conversations. The artificial data has been used in this study. So that the direct exposure of the sensitive data can be reduce as much as possible. If this system will be used in real life though there will be needed reducing the data, manage the accessibility, protected storage, Explicit preservation and removal policies, login of the audit and secure of the safety of vulnerable data from the beginning to the end of the whole process. This study can does not claim that it follows any legal structure. This is not tested in this study. This is out of the scope f the study.

4.3.5 Cultural Generalisation

In the sector of social engineering, Language, Social relationship, culture, how people see the authority all of these things are the most important things. So the AI model based on the information of particular artificial peoples or the country may not work as the same on the other place this should not be assumed without any proof. In the context of Bangladesh, This a significant matter of concern to establish a system. On some occasions the COVA-X corpus represents the different things from the native languages, banking practices, social relationships and the authority expressing system. In the future information will work well to build a larger culture and different language, side by side it will maintain ethics and safety.

4.3.6 Environmental Sustainability and Deployment

Machine learning systems needs real computer related resources, like training, storage in the computer, inference and the process of repeated experiments. Significantly there will be needed more memory and computation than the classical TF-IDF and XGBoost pipeline is used here for Transformer finetuning. The project did not measure the consumption of the electricity, direct use of carbon. So there is not present any numerical environmental

benefit claims over the one architecture over another. But in the real life to establish this system layer strategy can be effective. The initial stream of alerts will handle by a lightweight detector and when those case detector can not verify that time there will be used more expensive review or human review. Figure 4.7: Proposed layered deployment strategy stated that the initials alerts are detected by the lightweight detector and when some cased this lightweight detector can not be sure that time there will be use of human review. This way the architecture could reduce the cost of processing a case and it will truly add value.

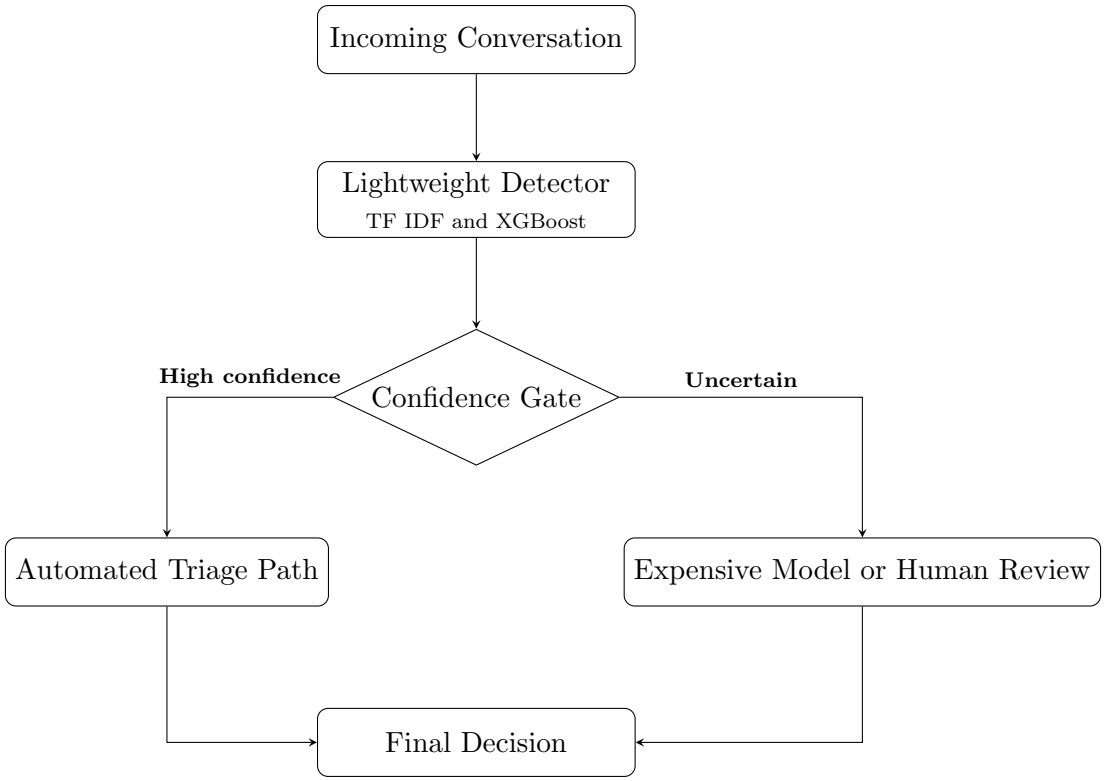


Figure 4.7: Proposed layered deployment strategy.

This type of design could reduce unnecessary computation by sending only uncertain cases to more expensive processing or human review. The routing threshold should still be determined using properly validated operational data, not directly from the current synthetic experiment.

The modular structure also makes future updates easier because the detector, calibration component, evidence layer, and Human AI interface can be improved separately.

4.4 Chapter Summary

The objective of this chapter was evaluating the conversational social engineering framework by using the executed notebook against the COVA-X v1.1 corpus. The audit name of the corpus come out framework-related differences between a few of the published de-

scriptions and the released data. The information generation-related can bring from the conversational text in the configuration mediation analysis. 73.61% accuracy landed by the XGBoost reproduction and 0.6835 of a macro F1. The context ablation showed just how much complied-class performance falls away when only the first six turns are visible, which underlines how difficult early triage really is. Longformer edged out DistilBERT in the reproduced transformer experiments, though neither configuration actually offered unrestricted full context. Temperature Scaling improved several calibration measures without touching classification accuracy. Two analyst-facing indicators survived into the evidence layer, though their measured faithfulness fell short of the threshold needed to call them model explanations outright. And the Human-AI simulation validated the planned statistical workflow, but it is no substitute for an actual participant study. Taken as a whole, this project makes the case for evaluating conversational detection alongside reproducibility, context availability, confidence reliability, evidence faithfulness, and HumanAI decision support, rather than treating any one of these in isolation. At the same time, the findings are fairly clear that the present implementation belongs in the category of an audited research framework — not a finished, autonomous security deployment ready for the field.

Chapter 5

Conclusion

5.1 Summary

This project examined whether multi turn conversational social engineering can be detected reliably enough to support security analysts. Using the COVA X v1.1 dataset, the study evaluated XGBoost with TF IDF, DistilBERT, and Longformer, while also examining context availability, confidence reliability, supporting evidence, and Human AI triage. The reproduced models performed below the published COVA X results. The experiments also showed that performance depends strongly on the available conversation context, especially for identifying eventual compliance. This makes the current system more suitable for retrospective classification than early triage. Post hoc calibration improved the probability estimates of the XGBoost model. Temperature Scaling improved calibration while preserving classification accuracy, while Vector Scaling improved several calibration measures with a small accuracy cost. The evidence layer retained banking information requests and financial requests as analyst facing corroborating evidence, but the faithfulness result did not support treating them as explanations of the model. The Human AI evaluation was tested through simulation rather than with real analysts. Therefore, the study validates the analysis framework but does not establish how the system would affect real analyst performance. Overall, the findings show that classification performance alone is not enough for a useful triage system. Context, confidence reliability, evidence quality, reproducibility, and human involvement all need to be considered. The proposed framework brings these components together, but it is not yet a complete real world SOC solution.

5.2 Limitations

Several limitations affect the interpretation of the results. The main limitation is the use of a synthetic dataset. COVA X allows controlled experimentation, but the generation process can introduce patterns that may not appear in real social engineering conversations. The configuration mediation results also show that scenario information can be

recovered from the generated text, which may affect model performance. The executed notebook used an 80/10/10 partition, while the later Version 2 methodology proposed 60/15/25. The reported results are valid for the executed experiment, but the alternative partition was not tested. Exact reproduction of the original COVA X baseline was also not possible because the published XGBoost system contains engineered features that are not sufficiently documented. The reproduced system therefore represents the recoverable part of the original pipeline. Some planned analyses were also incomplete. The early context experiment was performed with XGBoost but not with DistilBERT, the speaker channel results did not include the complete numerical breakdown, and the evidence and calibration analyses did not include every component planned in Version 2. Finally, the Human AI evaluation used simulated participant records instead of real security analysts. The study therefore makes no claim about real analyst accuracy, decision time, workload, or appropriate reliance. These limitations mean that the current system should be treated as a research framework and not as an operational autonomous security system.

5.3 Future Improvement

Future work should first complete the methodological gaps identified in this study. The intended 60/15/25 partition should be tested, the remaining evidence screening statistics should be reported, and the planned calibration analysis should be completed. Repeating the experiments under the revised methodology would also help determine which findings remain stable. Early prediction should become the main focus. The context results indicate that important classification information appears late in the conversation. Future models should therefore estimate escalation risk while the interaction is still developing. Sequential models, turn level updates, and methods for representing changes in attacker intent and victim response could be explored. The missing DistilBERT early context experiment should also be completed, along with the full numerical speaker channel analysis. This would help determine whether useful information comes mainly from attacker messages, victim responses, or their interaction. The evidence layer should use a fully reproducible quantitative screening process. The faithfulness analysis can also be extended so that useful analyst support is kept separate from unsupported claims about model reasoning. Real human participants should be included in the next stage. The four Human AI conditions can be tested with security analysts or suitably trained participants, with measures such as decision time, workload, confidence, and appropriate reliance. The scam type base rate condition should be retained to separate the effect of AI assistance from the effect of additional contextual information. Finally, future work should evaluate the framework on more realistic conversational data and consider richer models such as retrieval based systems, dynamic conversation modelling, and multimodal approaches. The modular structure should be maintained so that the detector, calibration layer, evidence layer, and Human AI interface can be improved separately. The main goal is not higher accuracy alone. The goal is a dependable triage system that provides useful and well

calibrated information to analysts when a decision needs to be made.

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